MIL-A-51356A(EA)
25 February 1987
SUPERSEDING
MIL-A-51356(MU)
9 April 1973

MILITARY SPECIFICATION

AREA PREDICTOR, RADIOLOGICAL FALLOUT, ABC-M5A2

This specification is approved for use within U.S. Army Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a radiological fallout area predictor, which is a transparent vinyl overlay for a map.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

PPP-T-42 - Tape, Packaging-Masking, Paper.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPT-S, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6665

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2.1.2 Drawings. The following drawings form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

DRAWINGS

124-13-27 - Area Predictor, Radiological Fallout, ABC M5A2.

SPECIAL PACKAGING INSTRUCTIONS (SPI)

P124-13-27 - Area Predictor, Radiological Fallout.

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. All materials cited on 124-13-27 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings. The ink used for marking and lettering shall be compatible with the vinyl chloride material. The marking and lettering shall be legible, non-fading, and durable.

3.2 Manufacture. The predictor shall be manufactured in accordance with Drawing 124-13-27.

3.3 Functioning. The predictor, when tested in accordance with 4.4.4.1, shall be capable of solving problems for downwind distance using data listed in table I.

3.4 Marking stability. The stability of the marking on the fallout area predictor shall be such that, when tested as specified in 4.4.4.2, not more than three characters (letters and numbers) shall have breaks or parts missing; not more than two breaks shall be present in any circle, semi-circle, or distance scale; not more than two tick marks on the azimuth or distance scale shall be partially obliterated; and no single tick mark on the azimuth or distance scale shall be completely missing.

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TABLE I. Problem data

Scale	Data			
	Problem 1	Problem 2	Problem 3	Problem 4
Yield - KT	1,000	400	20	0.2
Effective wind speed - KM/Hr	55	21	20	40
Downwind distance - KM	225 ± 10	90 ± 5	22 ± 2	2.2 ± 0.2

3.5 Dimensional accuracy. No part of the 40° lateral limit pattern or the six yield group circles, 1:250,000 scale, on the end item shall deviate from the patterns on the mylar (Drawing 124-13-28) by more than 1 millimeter and no part of the 40° lateral limit pattern or the six yield group circles, 1:50,000 scale, on the end item shall deviate from the patterns on the mylar (Drawing 124-13-28) by more than 2 millimeters when tested as specified in 4.4.4.3.

3.6 Dimensional accuracy after marking stability testing. After testing for marking stability in accordance with 4.4.4.2, the end item shall meet the requirements of 3.5 except that in the event of expansion of the end item the limits are changed to 2 millimeters on the 1:250,000 scale portion and 4 millimeters on the 1:50,000 scale portion.

3.7 Preproduction. Prior to the start of regular production, a preproduction sample of ABC-M5A2 area predictors shall be produced in accordance with this specification for examination and tests (see 4.3).

3.8 Workmanship. The predictor shall be free from imperfections such as cuts, tears, punctures, cracks, blisters, bumps, or dents and contamination such as grease, oil, or foreign matter.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not

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relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.2 Classification of inspection.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection

4.3.1 Lot. A preproduction lot of five M5A2 fallout area predictors shall be produced using the same methods, materials, equipment and processes as will be used in regular production.

4.3.2 Inspection procedures for examination and nondestructive tests. The preproduction lot of area predictors shall be examined for all the requirements of the drawings and this specification and shall be tested as specified in 4.4.4.

4.3.3 Acceptance/rejection criteria. The preproduction lot shall comply with the examination and tests specified in 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the fallout area predictors produced by one manufacturer, at one plant, from the same materials and under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For major defect 107 level S-4 shall be used, and for Major defect 108 level S-2 shall be used.

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample area predictors and the level A packaging shall be examined and tested in accordance with the classification of defects and table MIL-STD-105.

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4.4.3.2 Classification of defects.

(a) Area predictor, radiological fallout, ABC-M5A2 (Drawing 124-13-27.

Categories	Defects	Acceptance standard
Critical:	None defined	
Major:	AQL 1.0 percent defective	
101	Marking or lettering incorrect, illegible, or missing	
102	Color of markings	
103	Lettering and marking not on back surface	
104	Overall length	
105	Overall width	
106	Workmanship (see 3.8)	
107	Functioning	4.4.4.1
	AQL 2.5 percent defective	
108	Marking stability	4.4.4.2
109	Dimensional accuracy as produced	4.4.4.3
110	Dimensional accuracy after stability testing	

(b) Packaging inspection (section 5).

Categories	Defects
Critical:	None defined
Major:	AQL 6.5 percent defective
101	Unit and intermediate packing incorrect or missing
102	Quantity per intermediate pack incorrect
103	Unit and intermediate, and shipping container incorrect
104	Packaging containers broken, torn, crushed or punctured
105	Marking missing, incorrect or illegible

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning. From the nomogram on the area predictor, solve the four problems using the data listed in table I. Align the yield data and effective wind speed with a straight edge. Read the downwind distance from ground zero on the middle scale and compare with table I.

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4.4.4.2 Marking stability. Submerge the area predictor in a suitable container of water maintained at 125° ± 5°F for 30 ± 2 minutes with the entire printed side of the predictor in contact with the water. Remove the predictor from the water and wipe dry. Place the predictor on a smooth hard surface with the printed side of the predictor up. Within four minutes after removal from the water, apply five 1/2-inch by 6-inch strips of masking tape, conforming to type II of PPP-T-42, at right angles to the respective scales and nomograph so that portions or both the numbers and letters are covered. Roll a 10-pound, 5-inch diameter, hard surfaced roller at approximately 12 inches per 10 seconds over the strips of tape three times. Double back the end of each strip of tape and slowly and evenly peel each strip from the area predictor.

4.4.4.3 Dimensional accuracy. Dimensional accuracy of line patterns on area predictors shall be checked by superimposing a 1:1 scale, dimensionally-stable mylar duplicate of Drawing 124-13-28 over the manufactured end items.

5. PACKAGING

5.1 Packaging. Packaging shall be as specified in SPI P124-13-27.

6. NOTES

6.1 Intended use. This predictor is intended for use in making an immediate estimate of the potential location of the radioactive fallout hazard following a nuclear burst.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of packing required
(c) Type of exterior container or optional
(d) Unitization required

6.3 Subject term (key word) listing.

Area predictor
Fallout
Hazard
Map overlay
Nuclear burst
Overlay
Predictor
Radioactivity
Radiological fallout
Yield

6.4 International interest. Certain provisions of this specification are the subject of international standardization agreement QSTAG 283. When amendment, revision, or cancellation of this specification is proposed which will

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affect or violate the international agreement concerned, the preparing activity shall take appropriate reconciliation action through international standardization channels including departmental standardization offices, if required.

6.5 Annealing of plastic film. The plastic film, L-P-375, used in making the area predictor has a dimensional instability of up to 7 percent permissible by specification. It has been found that this instability can be reduced to an acceptable level by annealing. Annealing can be accomplished by heating flat sheets of the material to a temperature of 212°F for a short period of time and then allowing it to cool back to room temperature. The material should not be under stress or handled while at elevated temperatures as this may induce further dimensional instability.

Custodian:

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International interest (see 6.4)

Preparing activity:

Army - EA

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-A-51356A(EA)	AREA PREDICTOR, RADIOLOGICAL FALLOUT, ABC-M5A2

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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